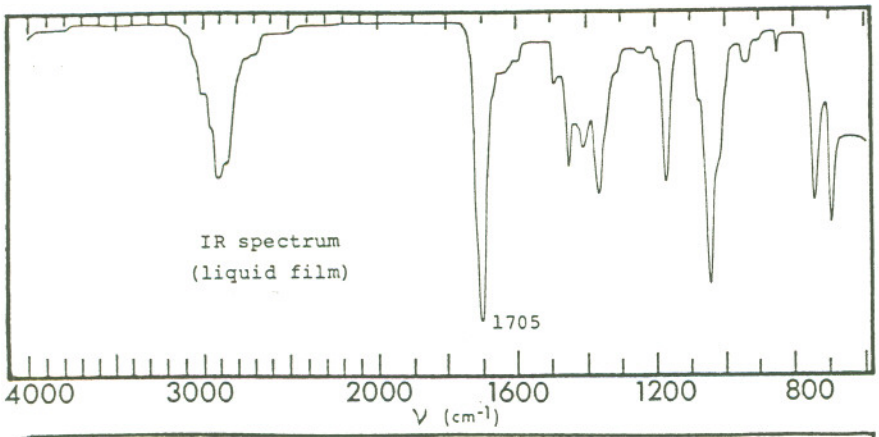
ORGANIC CUMULATIVE EXAMINATION

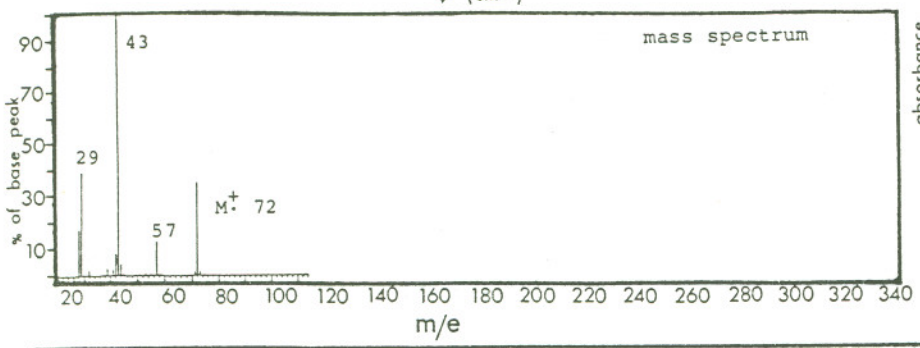
October 6, 1990

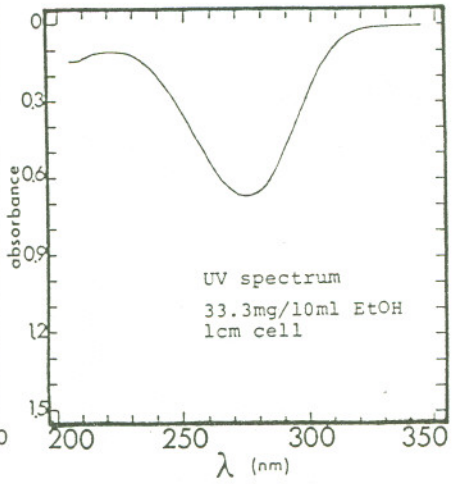
Assign structures for the following sets of spectra. Justify your answers by indicating important spectral features. Problems 1, 2, 10, & 11:

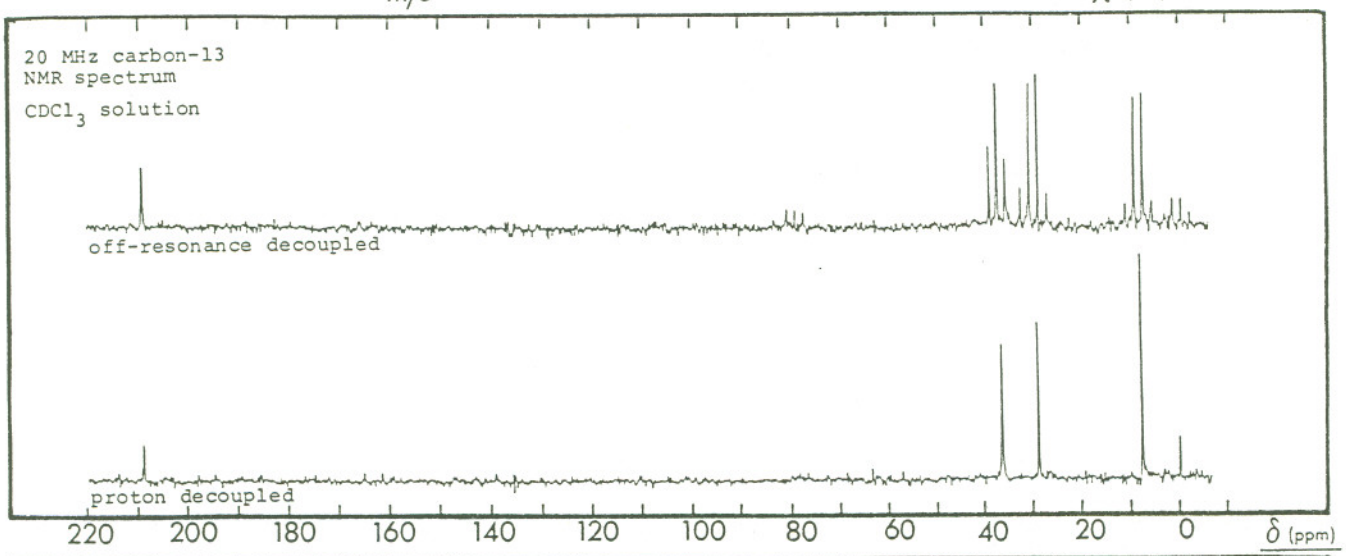
Answers will be posted on organic bulletin board next week.

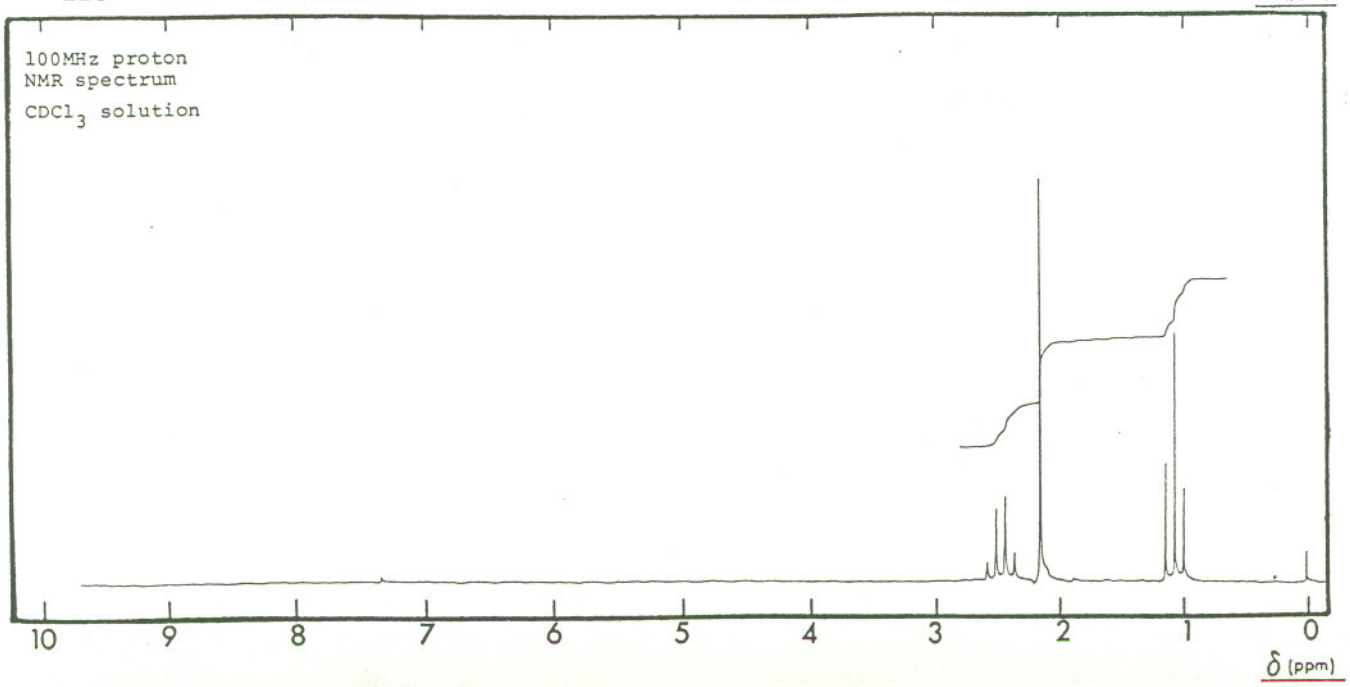
problem 1


IR spectrum
(liquid film)
1705
4000
3000
2000
1600
1200
800
ν (cm⁻¹)

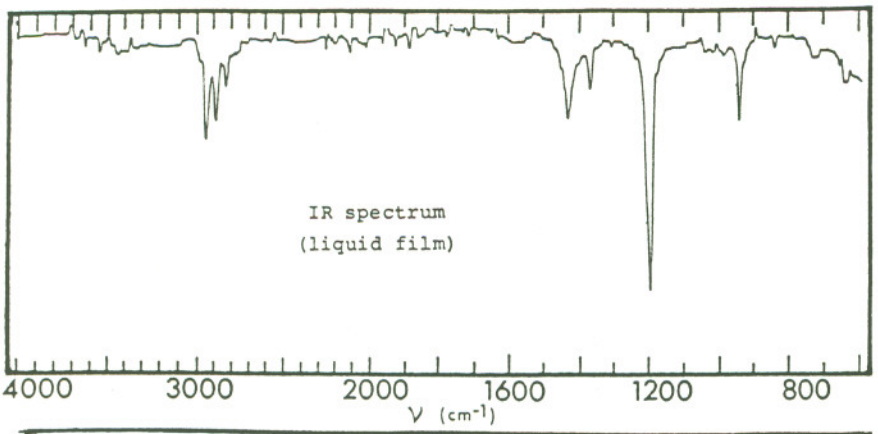

mass spectrum
% of base peak
90
70
50
30
10
43
29
57
M⁺ 72
20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340
m/e

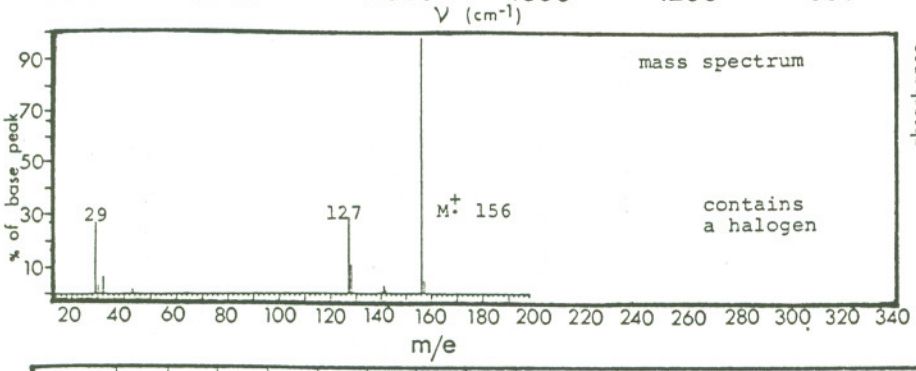

absorbance
0
0.3
0.6
0.9
1.2
1.5
UV spectrum
33.3mg/10ml EtOH
1cm cell
200
250
300
350
λ (nm)

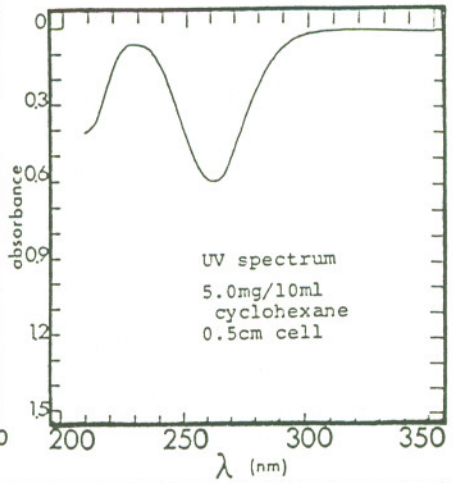

20 MHz carbon-13
NMR spectrum
CDCl3 solution
off-resonance decoupled
proton decoupled
220
200
180
160
140
120
100
80
60
40
20
0
δ (ppm)

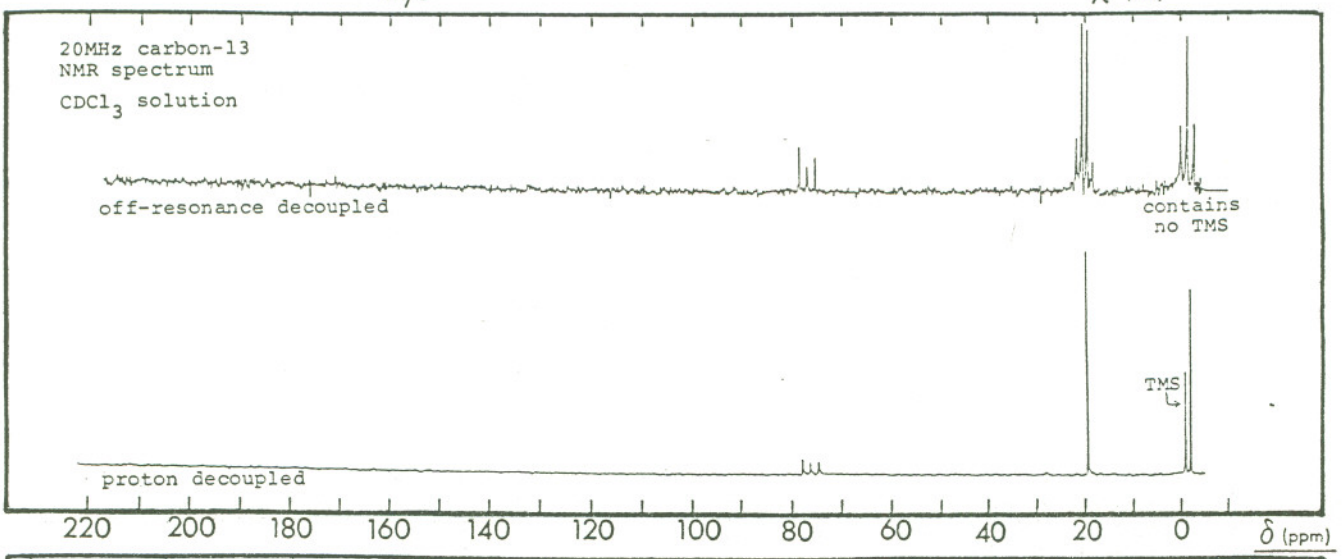

100MHz proton
NMR spectrum
CDCl3 solution
10
9
8
7
6
5
4
3
2
1
0
δ (ppm)

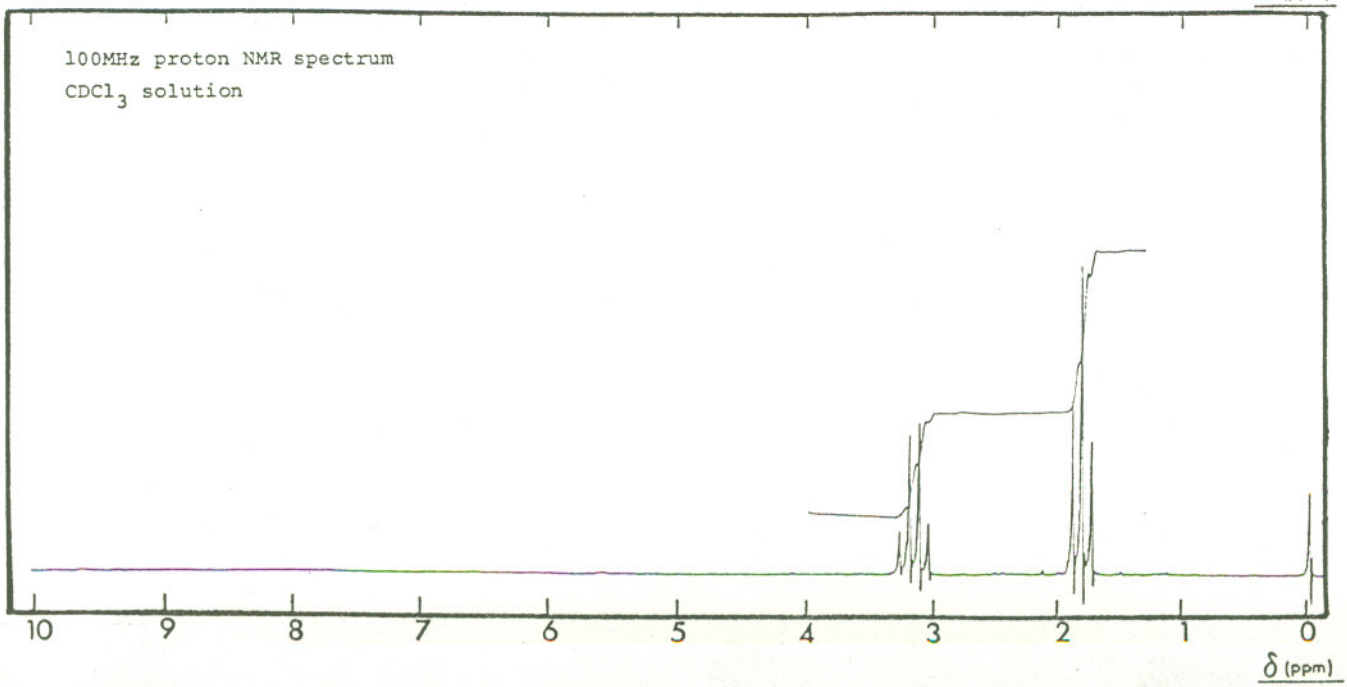
problem 2


IR spectrum
(liquid film)
4000
3000
2000
1600
1200
800
ν (cm⁻¹)

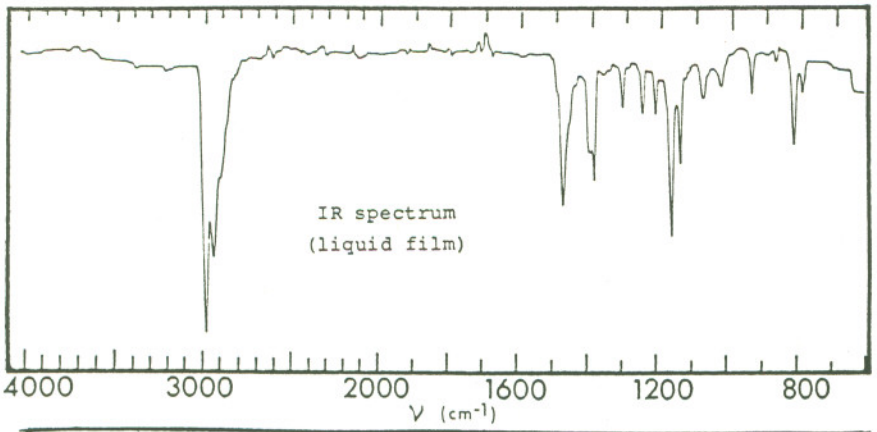

mass spectrum
% of base peak
90
70
50
30
10
29
127
M⁺ 156
contains
a halogen
20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340
m/e

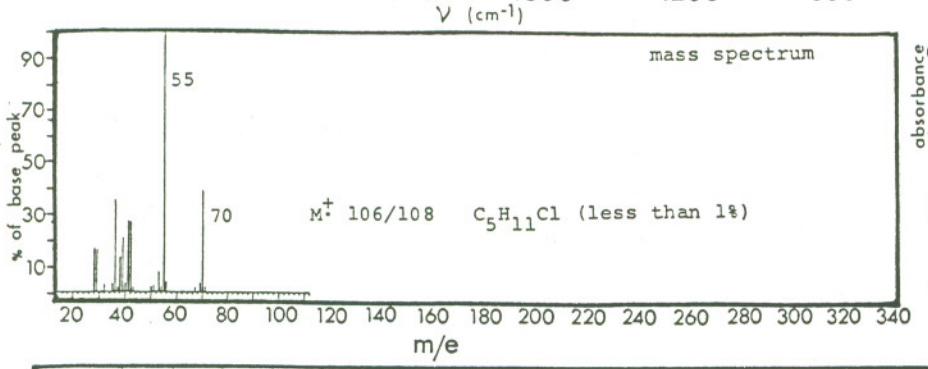

absorbance
0
0.3
0.6
0.9
1.2
1.5
UV spectrum
5.0mg/10ml
cyclohexane
0.5cm cell
200
250
300
350
λ (nm)

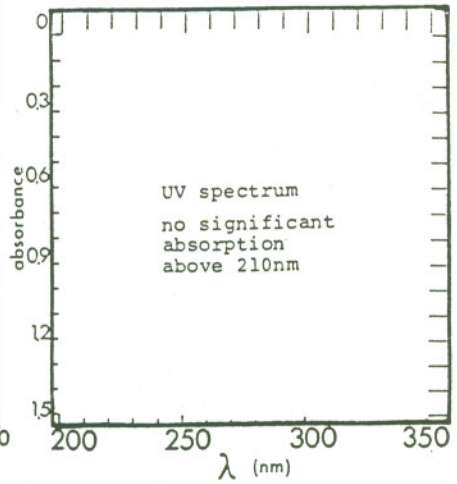

20MHz carbon-13
NMR spectrum
CDCl3 solution
off-resonance decoupled
contains
no TMS
TMS
proton decoupled
220
200
180
160
140
120
100
80
60
40
20
0
δ (ppm)

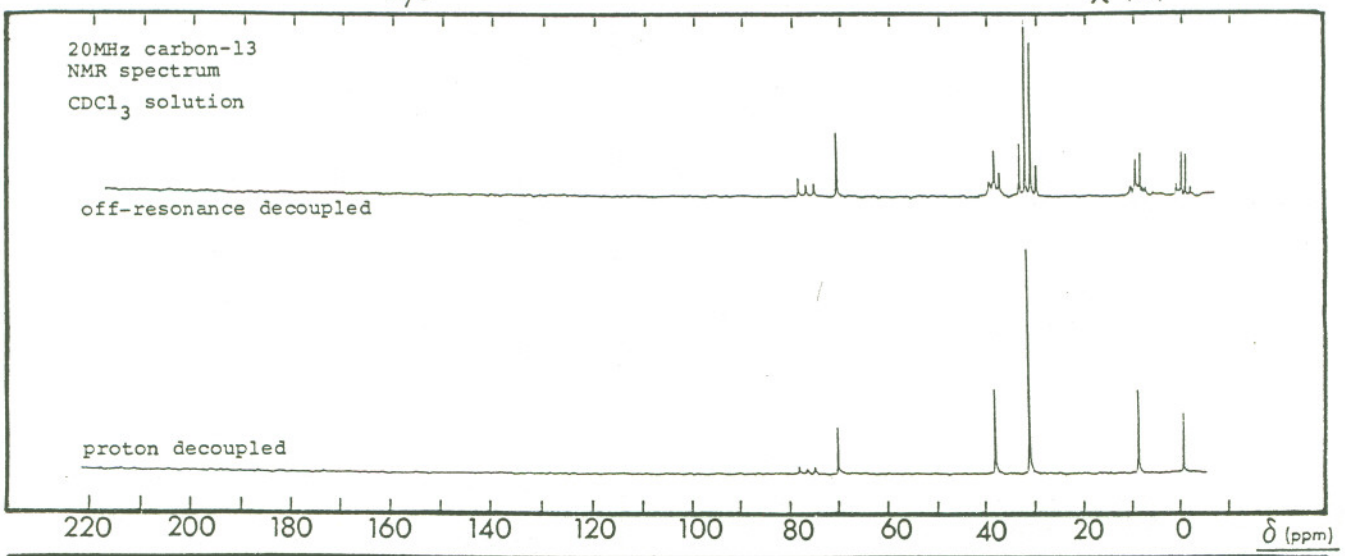

100MHz proton NMR spectrum
CDCl3 solution
10
9
8
7
6
5
4
3
2
1
0
δ (ppm)

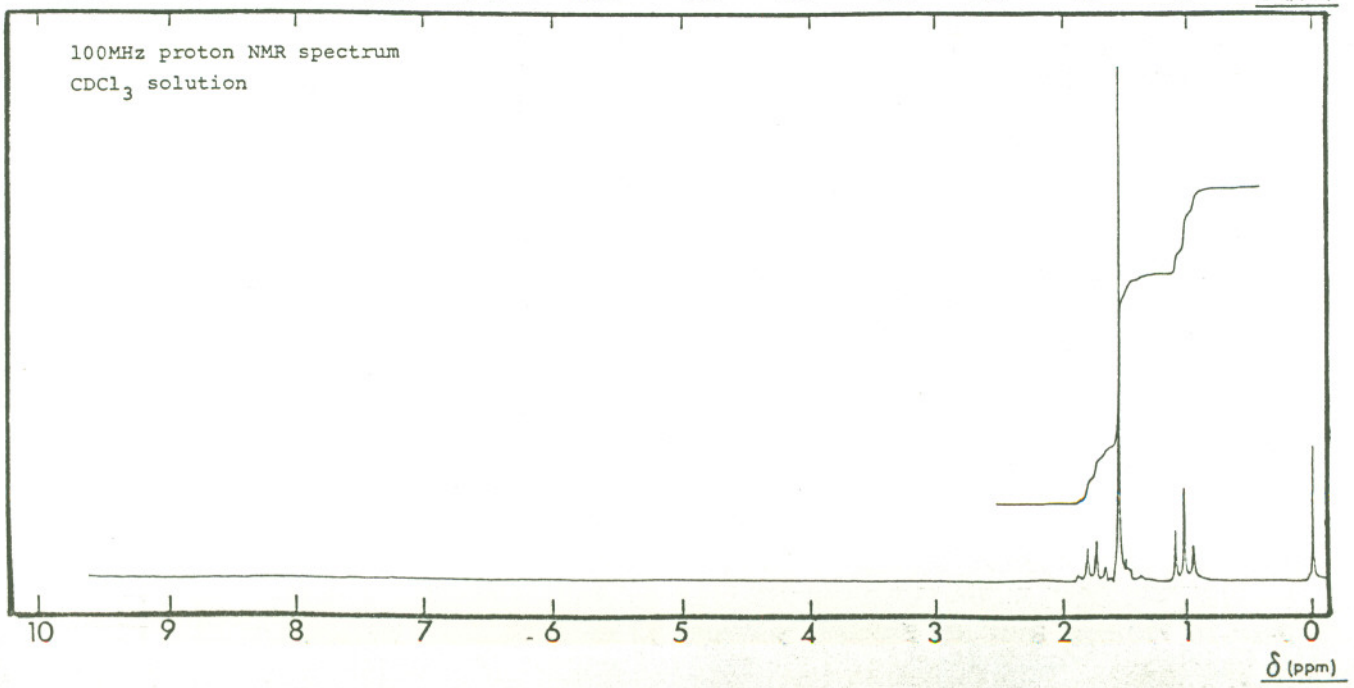
problem 10


IR spectrum
(liquid film)
4000
3000
2000
1600
1200
800
ν (cm⁻¹)

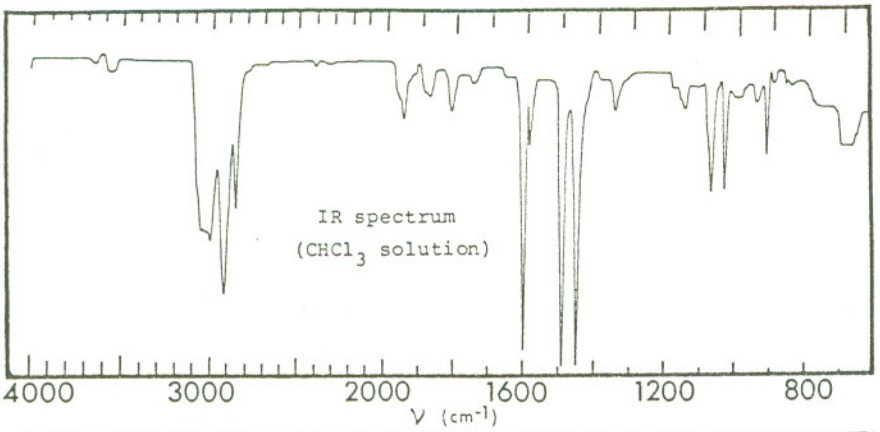

mass spectrum
55
70
M⁺· 106/108 C5H11Cl (less than 1%)
% of base peak
90
70
50
30
10
20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340
m/e

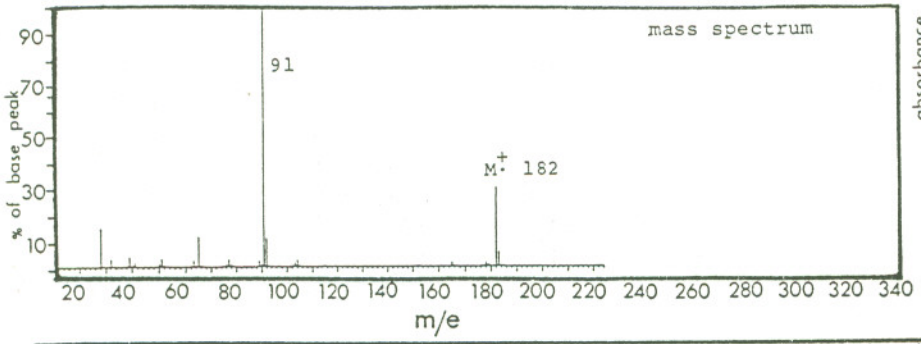

UV spectrum
no significant absorption above 210nm
absorbance
0
0.3
0.6
0.9
1.2
1.5
200
250
300
350
λ (nm)

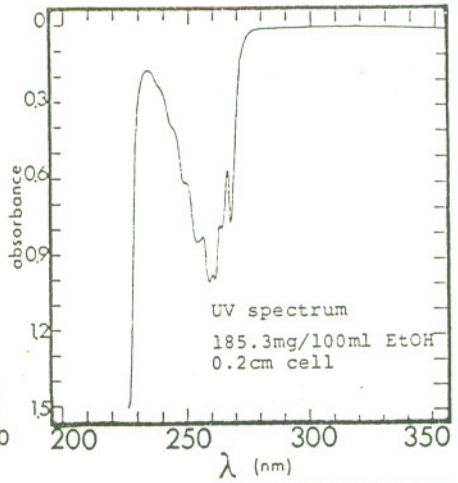

20MHz carbon-13
NMR spectrum
CDCl3 solution
off-resonance decoupled
proton decoupled
220 200 180 160 140 120 100 80 60 40 20 0
δ (ppm)

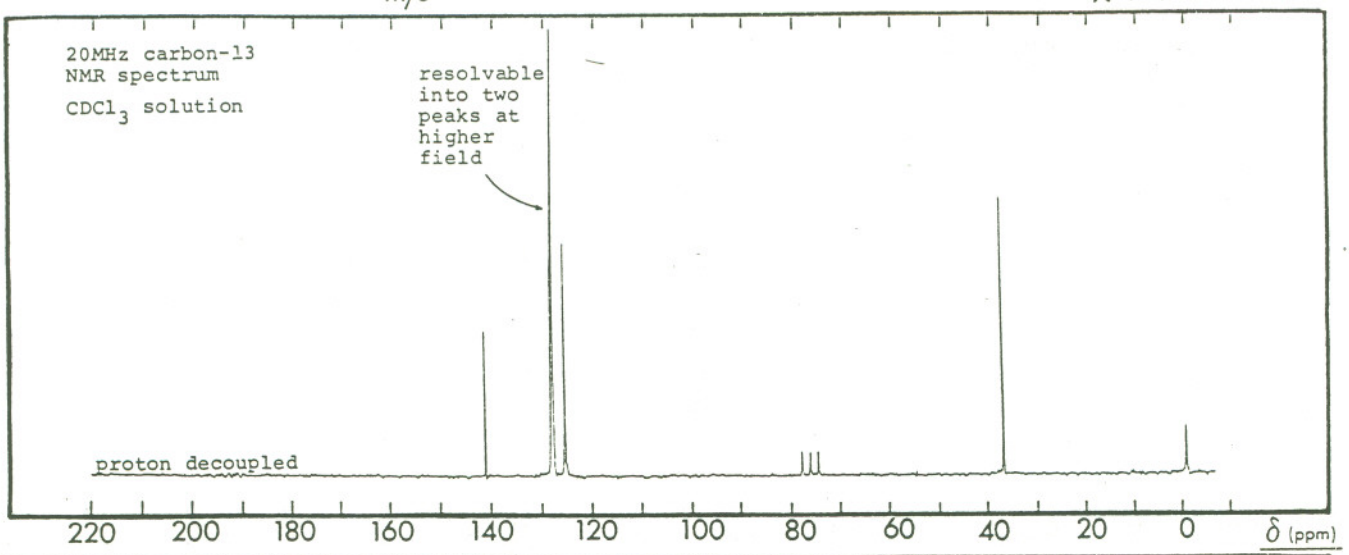

100MHz proton NMR spectrum
CDCl3 solution
10 9 8 7 6 5 4 3 2 1 0
δ (ppm)

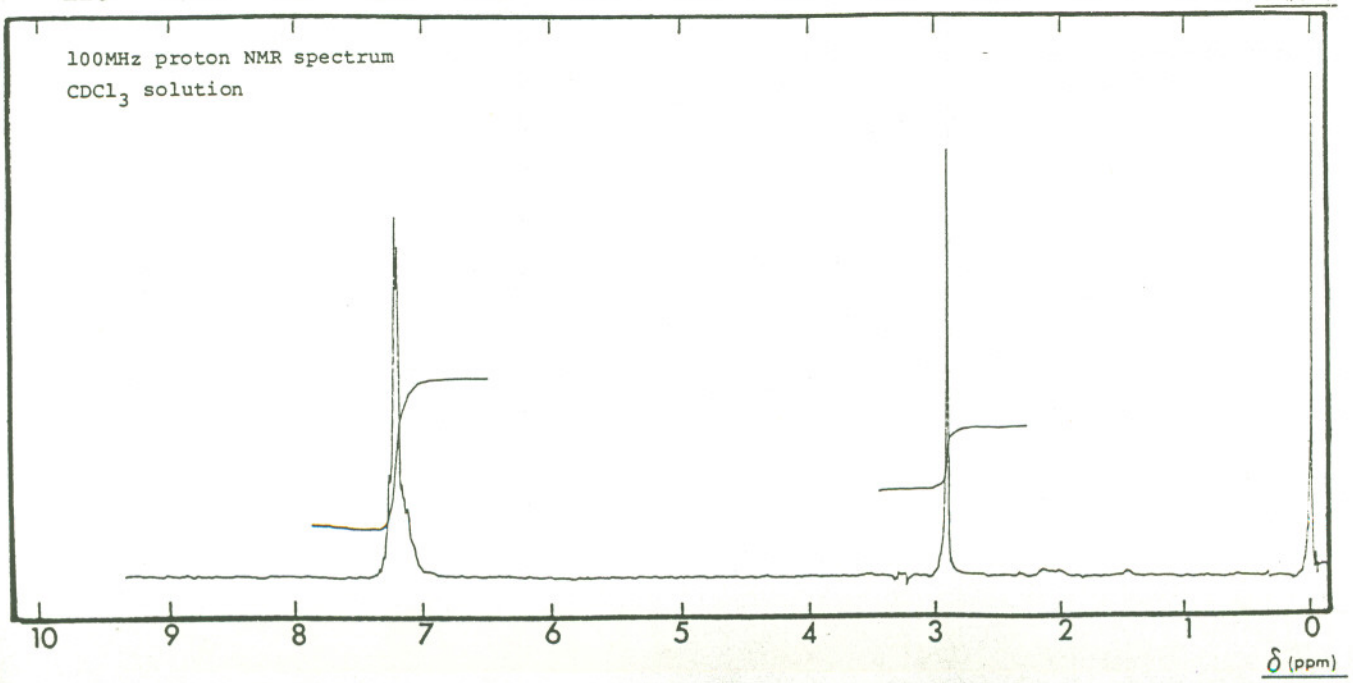
problem **11**


IR spectrum
($CHCl_3$ solution)
4000
3000
2000
1600
1200
800
ν (cm⁻¹)


mass spectrum
91
M⁺ 182
% of base peak
90
70
50
30
10
20 40 60 80 100 120 140 160 180 200 220 240 260 280 300 320 340
m/e


UV spectrum
185.3mg/100ml EtOH
0.2cm cell
absorbance
0
0.3
0.6
0.9
1.2
1.5
200
250
300
350
λ (nm)


20MHz carbon-13
NMR spectrum
$CDCl_3$ solution
resolvable
into two
peaks at
higher
field
proton decoupled
220 200 180 160 140 120 100 80 60 40 20 0
δ (ppm)


100MHz proton NMR spectrum
$CDCl_3$ solution
10 9 8 7 6 5 4 3 2 1 0
δ (ppm)